# AADIM NEPAL

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## EDUCATION

**New York University Abu Dhabi** 2022–2026
Bachelor's Degree in Mathematics and Computer Science cGPA: **3.875**

*Relevant Coursework:* Honors Analysis, Linear and Nonlinear Optimization, Deep Learning (Graduate), Foundations of Machine Learning (Graduate), Information Theory for Deep Learning, Graph Machine Learning, Stochastic Processes

## EXPERIENCE

**New York University Abu Dhabi** Ongoing
*Junior Research Scientist* *The Deep Learning Lab*

· Advised by Prof. Keith Ross and working in close collaboration with Prof. Yilun Du at Harvard University.

- Exploring a modular formulation of reinforcement learning in which different components of the decision process are represented by distinct learned energy functions.
- Studying how these energy functions can be updated from experience and composed at test time, including through mixture-based composition, to guide trajectory-level planning, support diverse plans and exploration, and adapt online as the model of the environment changes.

**The Deep Learning Lab** January 2025 – Present
*Research Assistant* *New York University Abu Dhabi*

· Supervisors: Prof. Keith Ross and Ravid Shwartz-Ziv

- Conducted research on LLM interpretability, studying how transformer layers contribute to mathematical reasoning and how their importance changes after fine-tuning and post-training.
- Fine-tuned language models using supervised distillation and reinforcement learning with verifiable rewards, with a focus on data-efficient approaches to improving reasoning.
- Contributed to papers accepted at the EMNLP 2025 Main Conference and workshops at NeurIPS, EMNLP, and RLC.

**Incubate Nepal** June 2026 – August 2026
*Technical Mentor* *Interpretable AI for Healthcare*

- Mentored a team of six students as they developed an interpretable AI platform for analyzing radiology scans.
- Advised the students on project scoping, model development, interpretability methods, and system integration; the students carried out the implementation, testing, and deployment.

**The Clinical AI Lab** September 2023 – August 2024
*Research Assistant* *New York University Abu Dhabi*

· Supervisor: Prof. Farah E. Shamout

- Developed and evaluated a multimodal deep-learning system combining retinal OCT scans with electronic health records from Cleveland Clinic Abu Dhabi to predict stroke risk.

- Implemented parts of the imaging pipeline, including preprocessing, feature extraction, multimodal fusion, model evaluation, and interpretability analysis. The resulting paper was accepted at **IEEE EMBC 2025**.

## PUBLICATIONS

- **A. Nepal**, Y. Du*, K. W. Ross*. Online Energy-Based Trajectory Planning. *Ongoing work, manuscript in preparation*, 2026.
- Y. Omori, Z. Dong, **A. Nepal**, A. Timsina, K. W. Ross. Decomposed Contrastive Critics for Continual Reinforcement Learning. *Continual Reinforcement Learning Workshop at RLC*, 2026.
- **A. Nepal**, S. Shrestha, A. Shrestha, M. Kim, K. Ross. Layer Importance for Mathematical Reasoning is Forged in Pre-Training and Invariant after Post-Training. *EMNLP BlackboxNLP Workshop and NeurIPS MATH-AI Workshop*, 2025. arXiv:2506.22638.
- M. Kim, A. Shrestha, S. Shrestha, **A. Nepal**, K. Ross. Reinforcement Learning vs. Distillation: Understanding Accuracy and Capability in LLM Reasoning. *NeurIPS MATH-AI Workshop*, 2025. arXiv:2505.14216.
- S. Shrestha, M. Kim, **A. Nepal**, A. Shrestha, K. Ross. Warm Up Before You Train: Unlocking General Reasoning in Resource-Constrained Settings. *EMNLP Main Conference*, 2025. arXiv:2505.13718.
- S. Shurrab, **A. Nepal**, T. J. John, N. G. Ghazi, B. Piechowski-Jozwiak, et al. Multimodal Deep Learning for Stroke Prediction and Detection using Retinal Imaging and Clinical Data. *IEEE EMBC*, 2025.

## AWARDS AND ACHIEVEMENTS

- **Slush'D Hackathon, NYU Abu Dhabi** 2024
  Placed among the top 10 teams after developing a full-stack educational platform. Worked on the React frontend and its integration with backend services, data storage, and learning workflows.
- **Yann LeCun's Graduate Deep Learning Competition** 2024
  Placed 5th as an undergraduate in a competition primarily for master's and PhD students. Developed a joint-embedding predictive architecture to predict snake movement across sequences of grid images.
- **International Biology Olympiad (IBO)** 2020
  Placed among the top four participants at the Nepal National Biology Olympiad and was selected to represent Nepal at the International Biology Olympiad.
- **World Schools Debating Championship (WSDC)** 2020
  Represented Nepal as a member of the National Debate Team. The team narrowly missed advancing to the elimination rounds.
- **NYUAD Swim Team** 2022–2026
  Member of the New York University Abu Dhabi Swim Team.
- **NYUAD Tennis Team** 2023–2026
  Member of the NYUAD Tennis Club's Advanced Tennis Team.